CASE STUDY I

IS3005 - STATISTICS IN PRACTICE I

SPEEDYCALL CUSTOMER CHURN DATA

Pasindu Punsara Jayasundara

s16806

Abstract

This study examines the factors that influence customer churn at a telecommunication company named SpeedyCall in USA.. The analysis mainly focuses on how contract type and payment method relate to whether a customer will churn or not. Since the study is descriptive in nature, the focus is on graphical summaries and basic statistical tests, without using any model fitting or predictive methods. Bar charts, Multiple bar chart and Percentage component bar chart were used to compare churn across different groups, and chi-square tests were carried out to check if the relationships were statistically significant.

The results show that customers with Month to month contracts , those using Electronic check have the highest churn rates. In contrast, customers with longer contracts and those using automatic payment methods are less likely to leave. The chi-square tests confirm that both contract type and payment method are strongly associated with churn.

Overall, the findings suggest that SpeedyCall can reduce churn by promoting longer-term contracts and encouraging customers to switch to automatic payment methods.

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Introduction

Customer retention has become a crucial challenge for telecommunication companies in a highly competitive market. When customers discontinue a service and move to another provider, it directly affects company revenue and growth. Understanding the factors that lead to customer churn allows companies to take preventive actions and improve customer satisfaction.

This case study focuses on SpeedyCall, a telecommunication company in the USA, which has collected various customer related data, including service details, billing information, and demographic characteristics. By studying these variables, the company can identify why customers discontinue their services and what actions can help reduce churn.

Objectives

To examine how **contract type** and **payment method** affect the customer churn among SpeedyCall customers.

1. To explore the distribution of customer churn across different contract types and payment methods using graphical summaries.
2. To test whether contract type and payment method are statistically associated with churn using Chi-square tests of independence.
3. To identify which customer groups have the highest risk of churn and provide recomandation to help SpeedyCall improve retention.

Data

In this study, the main interest is to examine how **contract type** and **payment method** influence the customer churn**. These two factors are strongly related to customer loyalty and payment convenience**, which are important in longterm service retention. The findings of this analysis will help SpeedyCall design better customer retention strategies, improve service plans, and reduce the overall churn rate.

Table 1: Table of variables

Variable Name	**Description**
Churn	Whether the customer churned or not (Yes or No)
PaymentMethod	The customer's payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card (automatic))
Contract	The contract term of the customer (Month-to-month, One year, Two year)

Data Analysis

Table 2: Tabel of churn

YES	NO
1869 (26.54%)	5174 (73.46%)

This table shows out of the total customers,**1,869** have churned, which is about **26.54%** of the total customers. The remaining **5,174** customers (**73.46%**) have stayed with the company. This shows that while most customers continue their service, a considerable portion has decided to leave.

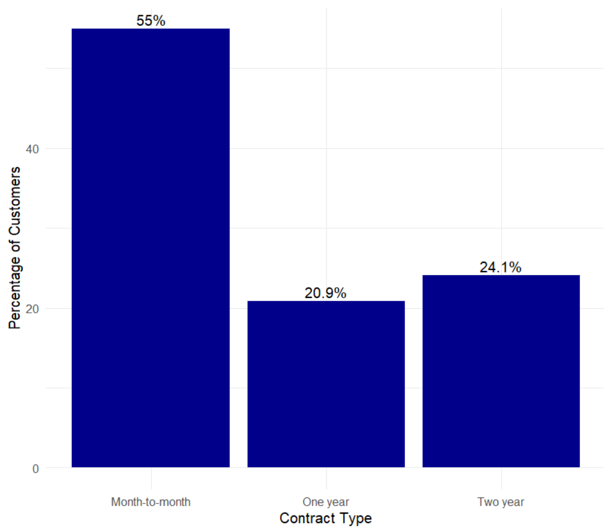


Figure 1: Bar chat of Contract Type

This bar graph shows how customers have chosen their contract type. More than half of the customers **(55%)** use a **Month-to-month** contract, making it the most common option. The **One-year** contract is chosen by **20.9%** of customers and **24.1%** have a **Two-year** contract. This means that the majority of customers prefer short term agreements, which may make it easier for them to leave the service compared to those with longer contracts.

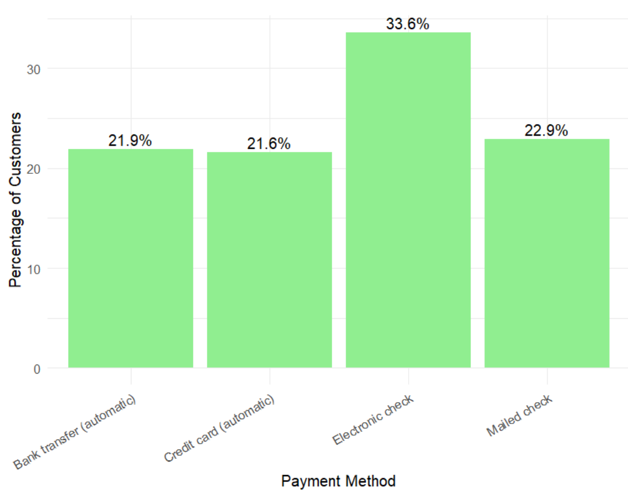
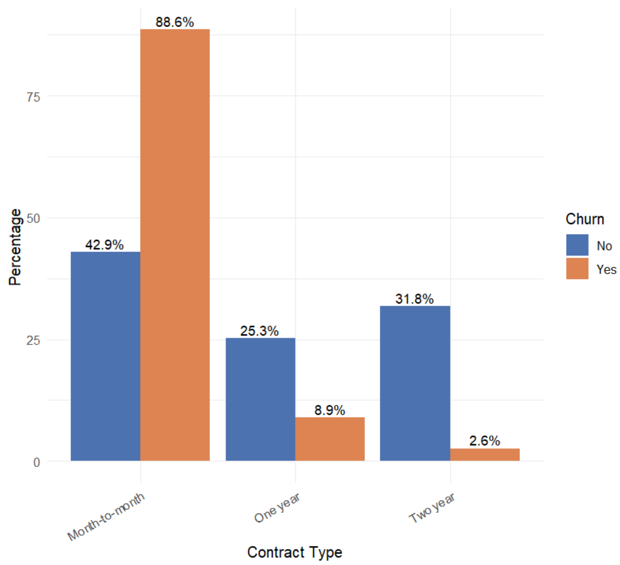


Figure 2: Bar chat of Payment Method

This bar graph shows the percentage of customers using each payment method. The most of the customers have chosen **Electronic check** method,that is **33.6%** of total customers. The other three methods have fairly similar usage: **Bank transfer (automatic)** at **21.9%**, **Credit card (automatic)** at **21.6%**, and **Mailed check** at **22.9%**. Since most of the customers use Electronic check method , patterns related to customer churn in this group can influence the overall results more than the other categories

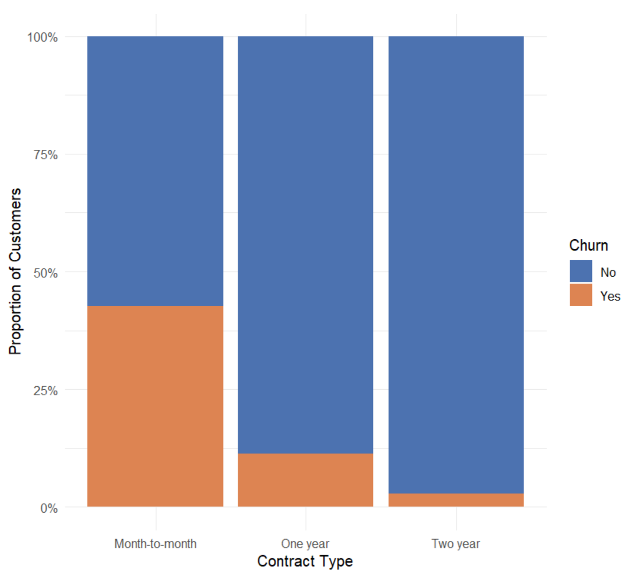



```
> table(Data$Contract, Data$Churn)

                  No  Yes
  Month-to-month 2220 1655
  One year       1307  166
  Two year       1647   48
```

Figure 3: Multiple bar chart of Contract type within each churn Group

This bar chart shows that most customers who churned (88.6%) were on Month-to-month contracts, while only 8.9% and 2.6% were on One-year and Two-year contracts, respectively. Since Month-to-month plans are flexible and used by most customers, they show the highest churn compared to longer contract commitments.

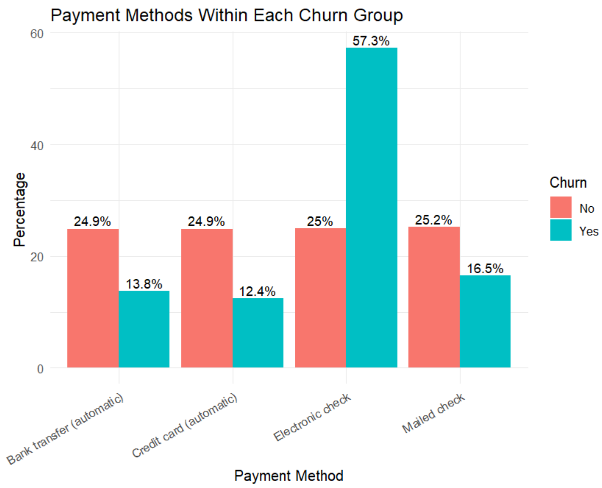



```
> percentages_Contract

                  No  Yes
  Month-to-month 57.3 42.7
  One year       88.7 11.3
  Two year       97.2  2.8
```

Figure 4: Percentage component bar chart of churn rate by Contract Type

This graph shows the churn rate for each contract type, where each contract group is taken as 100%. The Month-to-month contract group has the highest churn rate, with about 42.7% of these customers leaving. The Two-year contract group has the lowest churn rate, showing that customers with longer contracts leave less often.

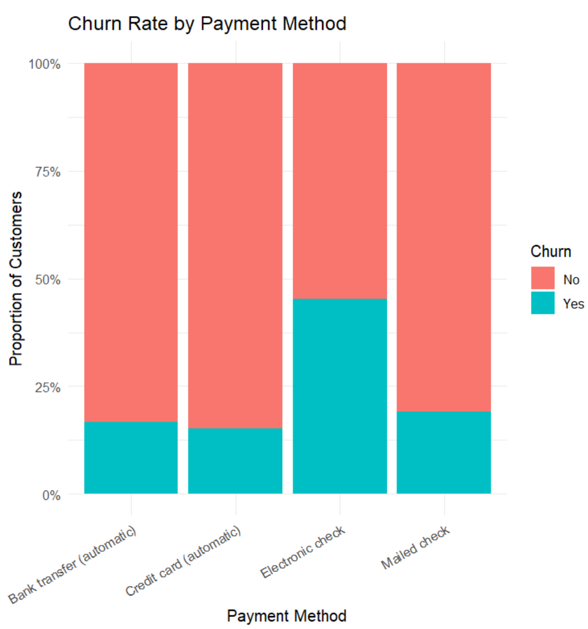



```
> table(Data$PaymentMethod, Data$Churn)

                              No  Yes
  Bank transfer (automatic) 1286  258
  Credit card (automatic)   1290  232
  Electronic check          1294 1071
  Mailed check              1304  308
```

Figure 5: Multiple bar chat of Payment Method within each churn Group

This graph shows the percentages of payment methods used by customers who stayed and customers who left. Electronic check has the highest churn, with 57.3% of churned customers using this method. Mailed check also has a noticeable churn percentage at 16.5%. In contrast, the two automatic payment methods show much lower churn: Bank transfer (automatic) at 13.8% and Credit card (automatic) at 12.4%. Both automatic methods have very similar and low churn rates.




```
> percentages_payments

                              No  Yes
  Bank transfer (automatic) 83.3 16.7
  Credit card (automatic)   84.8 15.2
  Electronic check          54.7 45.3
  Mailed check              80.9 19.1
```

Figure 6: Percentage component bar chart of churn rate by Payment method

This graph shows the churn rate for each payment method, where each payment method group is taken as 100%. Electronic check has the highest churn rate, with 45.3% of these customers leaving. Mailed check has a churn rate of 19.1%, which is higher than the automatic payment methods. Bank transfer (automatic) and credit card (automatic) both have much lower churn, at 16.7% and 15.2%, and they show very similar results. Overall, customers using automatic payment methods churn less than those using manual payment methods.

Now we will check if the relationships seen in the graphs are statistically significant.

Chi-square tests For Contract Type:

- H_0: There is no association between contract type and customer churn.
- H_1: There is an association between contract type and customer churn.

```
        Pearson's Chi-squared test

data:  counts_Contract
X-squared = 1184.6, df = 2, p-value < 2.2e-16
```

Since p-value < 0.05 reject Ho at 5% significant level. So there is an association between contract type and customer churn.

Chi-square For Payment Method:

- H_0: There is no association between payment method and customer churn.
- H_1: There is an association between payment method and customer churn.

```
        Pearson's Chi-squared test

data:  counts_payment
X-squared = 648.14, df = 3, p-value < 2.2e-16
```

Since p-value < 0.05 reject Ho 5% significant level. So there is an association between payment methods and customer churn.

General Discussion and Conclusion

- The analysis shows that Customer churn differs clearly by contract type. Month-to-month customers have the highest churn rate, while one-year and two-year contract customers leave less often. This suggests that longer contracts help keep customers with the company.

- Payment method also shows clear patterns. Customers who use electronic check have the highest churn rate, while those who use automatic payment methods (bank transfer or credit card) have the lowest churn rates. This indicates that customers who use automatic payments are more stable and less likely to cancel their service.

- The chi-square tests support the patterns seen in the graphs. Both contract type and payment method have statistically significant associations with customer churn. This means that these two factors genuinely influence whether customers stay or leave.

- The results are important for SpeedyCall because they highlight which customer groups are at higher risk of leaving. Month-to-month customers and electronic-check users are the groups that need the most attention.

- Based on the findings, SpeedyCall can reduce churn **by encouraging customers to switch to longer contract plans and promoting automatic payment methods**. Providing discounts, loyalty benefits, or easier upgrade options may help increase retention.

- Overall, understanding these patterns helps SpeedyCall design better retention strategies and improve customer satisfaction.